Chapter 6 Looking after Hunter 16

Hunter 16 is very robust and has been designed for a long and trouble-free useful life. However, like any precision instrument, it will only give its best if treated with a certain amount of care.

This chapter summarises the few simple rules that you should follow to ensure that **Hunter 16** gives you many years of trouble-free service.

6.1 Everyday Use

In everyday use, all you need to do is treat **Hunter 16** with a little respect:

- press the keys gently but firmly
- don't lay heavy items on top of it
- don't leave the display exposed to bright sunlight for long periods
- if a low power warning is displayed, turn **Hunter 16** off and renew or recharge the batteries as soon as convenient

Hunter 16 is designed for operation in normal indoor and outdoor environments. For more details, see Appendix A.

6.2 Case Sealing

All Hunter 16s are built with integral seals to provide a high degree of protection against atmospheric moisture, corrosive gases and accidental immersion, therefore **Hunter 16** should not be opened.

The battery seal is a small ‘O’ ring seal running in a machined groove in the battery plug and seating on a machined face in the **Hunter 16** casting. It requires a smear of silicon grease to assist in seating.

As the case is sealed, under certain circumstances (change of altitude, etc.) the interior of **Hunter 16** may develop a substantial pressure difference compared with the environment.

If there is internal overpressure, the keys will become stiff and difficult to operate. If there is external overpresure, **Hunter 16** will show a tendency to ingest any rainwater adjacent to the seals, and in extreme cases, may activate its keys.

In either case, release any pressure differential by loosening the battery cap.

If you are taking **Hunter 16** on an airline trip, this is a sensible precaution to take.

6.3 Humidity Indicator

Hunter 16's interior must be kept dry for correct operation. This is normally achieved by the silica gel dessicant which is included during manufacture.

The internal humidity is indicated by a humidity indicator in the top left corner of the display window. Check the colour of the indicator at intervals.

THE HUMIDITY INDICATOR MUST ALWAYS BE BLUE

If it takes on a pinkish their colour, local the unit must be returned to Husky Computers Ltd (or their local agents) promptly to avoid corrosion of internal parts. If the colour changes immediately after an accident involving water or high humidity, a fault has occurred in the sealing, and there could be water inside the unit.

If you suspect that water has entered the interior, you MUST take the following action:

1 Remove the batteries.

2 Return **Hunter 16** to HUSKY Computers, or authorised agent.

6.4 Storing Hunter 16

If storing **Hunter 16**, observe the storage conditions described in Appendix A.

If you wish **Hunter 16** to retain the data stored on its internal disk store it with its batteries fitted and check these at regular intervals (typically once a month), for charge and any signs of leakage. If the batteries are deteriorating, replace or recharge them at once.

6.5 Power

6.5.1 Introduction

Hunter 16 can be operated from three different power sources:

1. The supplied AC adaptor
2. Alkaline batteries
3. Rechargeable NiCad batteries

This ability to switch between these three power sources provides unrivalled convenience of operation. For example, you can power **Hunter 16** from the AC adaptor while in your office and fit it with rechargeable batteries for use while on the move. If you forget (or are unable) to recharge the batteries while away from the office, you can substitute easily obtainable alkaline batteries if required.

While NiCads can be recharged using the AC adaptor, alkaline batteries must not. To minimise the risk of attempting to recharge alkaline batteries, Hunter 16 can be set up to ask the user if charging is still allowed, the next time it is turned on after changing batteries. Applications can also disable charging and only enable it when appropriate (eg when requested by authorised personnel).

To increase battery life, special low power components and an automatic power-saving system are used. Power monitoring and warning systems are provided to keep you informed.

To protect the disk contents, a trickle-charged internal battery is fitted internally, which provides sufficient backup power to preserve the contents of the disk for at least a month, if the main power source is removed.

6.5.2 Maximising operating time

Hunter 16 will normally run for up to 35 hours on fresh Alkaline batteries when running a typical application, somewhat less on fully charged NiCads.

If it is important to you to gain the maximum operating time between battery changes/charges, follow these guidelines:

- Charge **Hunter 16** fully before using it, or fit fresh batteries.

- Turn **Hunter 16** off when you are not using it. Remember that when you turn on again, the program you were running will continue exactly where you turned off, so there is no point in leaving **Hunter 16** turned on unless you are using it.
- Do not connect devices to **Hunter 16** unless you are going to use them, and disconnect them afterwards.
- Keep use of the backlight to a minimum.
- Use programs which have been written specifically for **Hunter 16** to use the power-saving facilities (see Part 3).

6.5.3 The power management system

Hunter 16 has a sophisticated power management system to provide efficient operation using batteries. The main points are as follows.

6.5.3.1 Data protection

As already mentioned, an internal rechargeable battery is fitted, which is trickle charged from the main power source and used to protect the contents of the RAM disk. This battery should not need replacing.

6.5.3.2 Power monitoring

Hunter 16 provides two alternative power monitoring modes. You may be allowed to select the mode (eg by using the HOT key), or it may be selected for you (eg by your application).

If **Advanced Power Mode** (APM) is enabled, **Hunter 16** will monitor the battery level continuously so it can be read by an application or displayed pressing the HOT key ([PAW][H]). The level is displayed as a percentage of total battery capacity. Access to charging depends on the charging authorisation level. For example, if level 3 is selected and you change batteries, **Hunter 16** will ask you whether charging is still allowed when you next turn it on.

If Advanced Power Mode is disabled, **Hunter 16** is in **Standard Power Mode** (SPM) so you can no longer check the battery level, but other aspects of power conservation and low power handling are similar.

6.5.3.3 Low Battery Warning

Hunter 16 is designed to provide adequate warning of impending battery failure and will protect user's data, even in the worst situations.

If Advanced Power Mode (APM) is enabled, you can set the battery level at which warning is given (normally, 5% charge) by using the HOT key.

Hunter 16 will also issue low power warnings if the battery voltage drops below 4.5V, showing that the batteries are almost exhausted. These warnings will be issued even if APM is disabled.

The low power warning message is displayed on the top line of the screen and looks like this:

```
WARNING BATTERY LEVEL L0W
```

Hunter 16 bleeps twice to draw attention to the warning, then restores the original display until the next warning.

Since it is usually essential that all data and programs are preserved, you should switch off as soon as practical and either replace the batteries with a fresh set or, if rechargeable batteries are fitted, put them on charge.

If you do not turn off, the warnings will be repeated every 5 seconds.

If you ignore ten consecutive warnings, **Hunter 16** will switch off automatically (unless this is overridden by your application).

When power is restored, program execution will continue from precisely where it was prior to the interruption.

If you attempt to use **Hunter 16** without replacing or recharging the batteries or connecting an AC adaptor, another warning message is

displayed and **Hunter 16** immediately turns off again.

Note that flat batteries can partially recover after being switched off, allowing operation for a short period. This is not good practice, particularly with rechargeable batteries, as these could become totally exhausted, damaging the cells permanently.

6.5.3.4 Auto Timeout

If **Hunter 16** is left turned on and not used, it will emit a warning bleep every 2.5 minutes to remind the user that it is turned on. To conserve battery life, it will switch off automatically if the keyboard is not touched for a period of 5 minutes (this period may be changed by an application program: see Part 3). When the user turns on again, execution will continue normally.

6.5.3.5 Power Save

An automatic power conservation system is incorporated which monitors operation and removes power from circuits when they are not in use. **Hunter 16** will enter power save mode automatically whenever it is waiting for keyboard input or serial input; this has no effect on operation other than prolonging battery life.

6.5.3.6 Charging authorisation

Hunter 16 can be used with Alkaline (non-rechargeable) or NiCad (rechargeable) batteries. NiCads can be charged by fitting the AC adaptor, but Alkalines must not, as they may explode, jam in the battery tube or leak corrosive chemicals.

If Advanced Power Mode is enabled **Hunter 16** uses a charging authorisation' mechanism to help prevent problems. By default, the authorisation level is set to 1, meaning that the AC adaptor will not charge the batteries.

While this is quite safe, it means you must remove the NiCads to recharge them, which may be inconvenient. To allow charging from the AC adaptor, you must change the authorisation and then enable charging, by either using the HOT key power option, or your application. You should be provided with access to one (or both) of these facilities if you are supplied with an AC adaptor and expected to recharge your own batteries.

If Advanced Power Mode is disabled, charging may be enabled or disabled (by your application). Your application may also provide a fast charge option.

If you are provided with an AC adaptor you MUST follow the instructions provided with your application concerning its use.

6.5.4 Using non-rechargeable batteries

THESE INSTRUCTIONS RELATE TO PERSONNEL SAFETY AS WELL AS RELIABLE OPERATION OF HUNTER 16 AND AC ADAPTOR. IT IS IMPORTANT THAT THEY ARE READ AND UNDERSTOOD.

We strongly recommend the use of Alkaline-Manganese batteries, such as Mallory MNI500s. You will need four AA size batteries to operate **Hunter 16**.

We do not recommend use of the cheaper Zinc-Carbon batteries for routine operation, as they may leak (damaging **Hunter 16**), have shorter lives, and may cause technical problems. You may however use such batteries for short periods, in an emergency.

Alkaline cells should provide up to 35 hours of operation when running a typical application, and a shelf life of approximately one month. There is no need to remove them from **Hunter 16** during shipment or storage, but remember that their capacity will decrease slowly with time.

6.5.4.1 When to change batteries

You should change the batteries if:

- The following message is displayed every few seconds when you turn **Hunter 16** on:

  `WARNING BATTERY LEVEL LOW`

- A message is displayed by the application telling you that power is low
- **Hunter 16** will not power up
- You know that the batteries are nearly exhausted and will be relying on the batteries for an extended period. (Typical Alkaline batteries provide up to 35 hours' continuous operation.)

You may, if you prefer, operate **Hunter 16** from an AC adaptor or change to NiCads, as described in subsequent sections.

NOTE: If the main batteries are removed or completely discharged, the contents of the RAM disk will still be preserved by another independent internal battery. This should protect the disk contents for at least four weeks, but does not allow operation of Hunter 16.

If Hunter 16 may be without power for more than four weeks, copy any files that you want to keep onto external disks or other media first, for added security.

6.5.4.2 How to change batteries

The batteries are loaded into the battery tube and held in place by a spring-loaded threaded plug. Install them as follows:

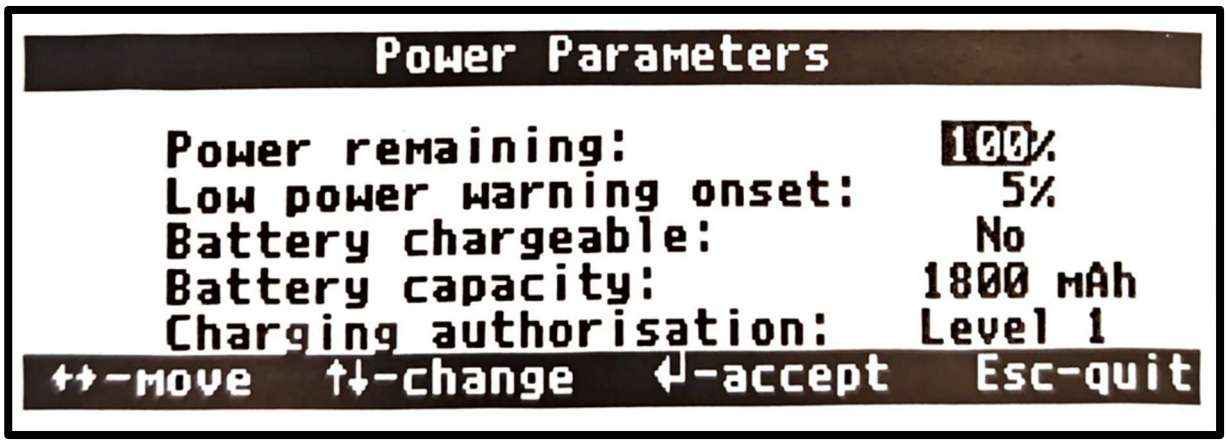
- Turn off.

- ONLY USE A COIN in the battery plug. Screwdrivers will damage the slot and **Hunter 16's** appearance.
- Remove the old batteries and dispose of them safely.
- Feed in the cells, positive inwards. Don't drop the batteries in vertically - they can be damaged. Instead, slope **Hunter 16** slightly.
- Take the battery plug and, with finger pressure only, press it into the battery compartment and turn it clockwise.
- Only when the thread is started, use a coin to screw the plug home. The battery plug should stand about 2mm proud from the case wall.

  NOTE: Turn the plug clockwise to insert it, anti-clockwise to remove it.

- Turn on and you should be able to carry on from where you left off using **Hunter 16** (assuming Advanced Power Mode is disabled).

OR, if Advanced Power Mode is enabled, when you turn on **Hunter 16** will display:



- Set the power level to 100% by using the cursor keys, then press O to carry on using Hunter 16.

If Advanced Power Mode is enabled but you do not see this screen, your application may provide a similar facility, with a name such as ‘New battery’. (If in doubt, ask your technical support team.)

NOTE: Do not mix batteries. Replace all four at the same time, with batteries of the same type.

6.5.4.3 Changing to NiCads

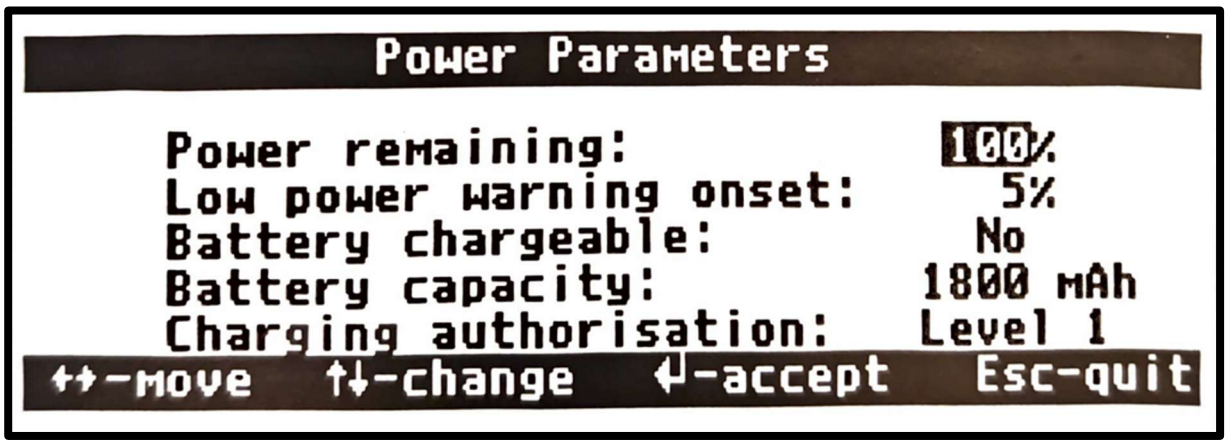
Hunter 16 can be operated from four standard AA size NiCad batteries. To change over to rechargeable batteries:

- Turn off.
- Remove the Alkaline batteries and fit the rechargeable batteries in their place, as described in section 6.5.4.2.

  NOTE: These batteries must be at least partly charged! (New NiCads are usually not charged.)

- Turn on and you should be able to carry on from where you left off using **Hunter 16** (assuming Advanced Power Mode is disabled). You may also need to enable charging: see your application's documentation.

OR, if Advanced Power Mode is enabled, when you turn on **Hunter 16** should display:



- The highlight should be next to the **`Power remaining`** prompt. If not, use [Right] or [Left] to move to it. Use [Up] or [Down] to adjust the **`Power remaining`** to the power in the batteries.
- Press [Right] twice to move to the **`Battery chargeable authorisation`** option, then press to set it to Yes.
- Press [Right] again to move to the **`Battery capacity`** option, then use [Up] and [Down] to set the capacity. Typical NiCads have a capacity of 600 to 800mAh.
- Press [Right] again to move to the **`Charging authorisation`** option, then us [Up] to set the level as follows:

1. means do not allow charging
2. means allow charging only until **Hunter 16** is powered down (for whatever reason).
3. means allow charging until the battery cap is removed (or all power is lost), then ask if charging is still allowed.
4. means allow charging permanently. Use this option with caution.

   (If you are not sure which option you want, select option 2 to enable charging just for now.)

- When everything is set correctly, press [RETURN] to carry on using **Hunter 16**. If the batteries are not fully charged, charge them as described in section 6.5.5.2.

If Advanced Power Mode is enabled but you do not see the above screen, your application may provide a similar facility, with a name such as 'Changing to NiCads'. (If in doubt, ask your technical support team.)

6.5.4.4 Using the AC adaptor

If you want to power **Hunter 16** from an AC adaptor, you must first replace the alkaline batteries with NiCads (see section 6.5.4.3).

See section 6.5.6 for details of using the AC adaptor.

WARNING: NEVER Connect the AC adaptor when alkaline batteries are installed and charging is enabled. They will NOT re-charge. Instead, they may explode, jam in the battery tube or leak corrosive chemicals.

6.5.5 Using rechargeable batteries

THESE INSTRUCTIONS RELATE TO PERSONNEL SAFETY AS WELL AS RELIABLE OPERATION OF HUNTER 16 AND AC ADAPTOR. IT IS IMPORTANT THAT THEY ARE READ AND UNDERSTOOD.

Hunter 16 can be fitted with four standard AA-size NiCad rechargeable batteries. These can be charged while fitted by connecting the supplied AC adaptor, or can be removed and recharged

independently, using a separate charger (eg a 'fast charger'). Note that charging is normally disabled when operating at extremes of temperature (outside 5 to 40 degrees Centigrade).

6.5.5.1 When to recharge

You should recharge the batteries if:

- The following message is displayed every few seconds when you turn Hunter 16 on:

  WARNING BATTERIES ARE LOW

- A message is displayed by the application telling you that power is low.
- **Hunter 16** will not power up.
- You know that the batteries are nearly exhausted and will be relying on the batteries for an extended period. You can check the battery level by using the HOT key ([PAW][H]).
- It hasn't been recharged for a month. NiCads self-discharge, especially at elevated temperatures.

If you need to use **Hunter 16** right away, you can change the batteries rather than recharge them straight away, or use the AC adaptor (see 6.5.6). If you change to non-rechargeable batteries, you MUST remove charger authorisation or disable charging; refer to section 6.5.5.4 for details.

6.5.5.2 Recharging batteries

You can recharge the batteries in three different ways:

- By removing the batteries and recharging them in a separate charger. This will not be described further here. When you fit the batteries again, set the power level to 100%.
- By connecting the AC adaptor, once you have enabled charging.
- By using the DC fast charger, once you have enabled charging. Refer to the instructions supplied with the charger.

The special Hunter 16 AC adaptor is the only unit which may be used to recharge the NiCad batteries when they are installed in **Hunter 16**. The use of an ordinary AC adaptor or battery eliminator may damage **Hunter 16** or the batteries.

If Advanced Power Mode (APM) is enabled, charging must be enabled and charging authorisation must be granted before batteries will recharge. You may be able to change these by using the HOT key **`Power`** option, or your application may control these for you. If you do not know, consult your technical support team.

If APM is disabled, charging should be enabled, but may be turned off by your application. Your application may also allow you to select fast or boost charging. If in doubt, ask your technical support team.

6.5.5.2.1 WARNINGS

When using the AC adaptor be absolutely certain that **Hunter 16** has four NiCad cells correctly aligned and no other type is present. Do not even mix NiCads of different age or state of charge, as this produces a risk of chemical leakage, gassing or explosion. Do not attempt charging at temperatures outside the range 5 to 40 degrees C.

The AC adaptor contains lethal voltages, so under no circumstances should it be opened, in any way tampered with, or used for any other purpose. It must, of course, be kept dry. There are no user serviceable items inside the case.

If you suspect that the AC adaptor is faulty, you may change the fuse inside the AC plug for a similar fuse not exceeding 2 amps rating. The AC adaptor is fully protected against continuous short circuits but if unlikely fault conditions arise which cause overheating of the AC adaptor, it will self-destruct, quietly and safely. The AC adaptor is double insulated, so an earth connection is not necessary.

6.5.5.2.2 Procedure

To recharge using the AC adaptor the following procedure is recommended:

1. Switch off **Hunter 16**.
2. Plug the AC adaptor into **Hunter 16's** 5-way connector, by aligning the red mark on the plug sleeve with the corresponding red mark on the FISCHER socket. Push the plug in until the outer sleeve is flush with the **Hunter 16** casing.
3. Plug the AC adaptor into the AC socket.
4. Switch on the AC socket.

5. Charging should begin. Wait for the batteries to charge fully. If APM is enabled you can check the power level by turning on and using the HOT key ([PAW][H]).

   If convenient, you may leave the batteries to charge overnight. You may also disconnect the AC adaptor before the batteries are fully charged if you need to.

   WARNING: Do not remove the batteries while recharging.

6. When charging is finished, turn off at the AC socket, then disconnect the AC adaptor.

You may turn on and use **Hunter 16** during charging, but then charging will take longer or, under extreme conditions (eg continuous use of the serial port), not take place at all.

As with all electrical apparatus, the AC adaptor should be disconnected from the AC when not in use. Only 240V AC (220 to 250V), 50 or 60 Hz is suitable for standard units. AC adaptors for alternative supplies, such as 110V, are available.

6.5.5.2.3 If batteries will not recharge

If batteries will not recharge, the most likely reason is that charging has not been enabled as described in section 6.5.4.3.

If Advanced Power Mode (APM) is disabled, your application may have disabled charging. Read your application's user instructions to see how to enable charging.

If APM is enabled, the charging authorisation level may have turned off charging. For example, if level 2 is selected **Hunter 16** will disable charging when it is turned off. To check, use the HOT key ie press [PAW][H] and select the **`Power`** option:

- If Battery charge able is set to No press [Right] to move to it and press [Up] to change it to Yes
- If **`Charging authorisation`** is set to **`Level 1`**, move to it and use [U] to enable charging by selecting one of the following levels:

1. to allow charging until turned off (eg by using the [PWR] key or changing the batteries).

2. to allow charging until the batteries are changed, then ask the user if charging should still be allowed.
3. to allow charging continuously.

When you have finished, press [RETURN] and then [Esc] to leave the HOT key.

If the HOT key does not work or the Power option is not displayed, your application may provide similar facilities. If you are not sure refer to the application's user instructions or ask your technical support team.

If the HOT key shows that the batteries are chargeable, but they still will not recharge, they may need replacing. See section 6.5.5.3 for details.

NOTE: Even if the main batteries are removed or completely discharged, the contents of the RAM disk will still be preserved by another independent battery which is fitted internally. This should protect the disk contents for several weeks, but does not allow operation of **Hunter 16**. If **Hunter 16** may be without power for more than four weeks, copy any files that you want to keep onto external disks or other media first, for added security.

6.5.5.2.4 If battery capacity drops

The capacity of rechargeable batteries should only decrease slightly during their normal working period of three to four years. If the capacity drops considerably, you should try putting the batteries through several full charge/discharge cycles: run **Hunter 16** until battery low warnings are given then charge it fully, then repeat this cycle two or three more times. If this does not restore the batteries' capacity, you should replace them.

6.5.5.3 Replacing rechargeable batteries

In normal use, rechargeable batteries should last for three or four years

before they need to be replaced, although their life is reduced by repeated or prolonged total discharge and by excessively high temperatures.

During the normal operating period, their capacity should only slightly decrease. If it drops considerably, you should try putting the

batteries through several full charge/discharge cycles as described in section 6.5.5.2.4. If this does not restore the batteries' capacity, you should replace them.

To replace the batteries, follow the instructions in section 6.5.4.2, then charge them fully as described in section 6.5.5.2.

NOTE: Several discharge/charge cycles are required before NiCad cells reach their peak capacity.

6.5.5.4 Changing to non- rechargeable batteries

To change over to Alkaline batteries:

1. Turn off, remove the rechargeable batteries and fit the Alkaline batteries in their place, as described in section 6.5.4.2.
2. Turn on. If Advanced Power Mode is disabled, this should return you to your application. Select the application option which selects Alkaline batteries or disables charging (see your application's user instructions).

OR, if Advanced Power Mode is enabled:

3. If charging authorisation is level 3, the following message will be displayed:

   `Level 3- Continue charging : No`

   If No is not displayed, press [Up] to display it, then press [RETURN].

4. If the Power Parameters menu is not already displayed:
   - Press [PAW][H]
   - Select the **`Power`** option
5. You must now use the **`Power Parameters`** menu to disable charging and set the battery capacity and power level:

- Set the **`Charging authorisation`** to 1 to disable charging. This is essential for safety reasons.
- Set the **`Battery capacity`**. (Typical alkalines have a capacity of 1400mAh.)
- Set the **`Power remaining`**. (Typically to 100%, if the batteries are fresh.)

- Press [RETURN] and then [Esc] to leave the menu.

If you cannot display the Power Parameters menu, your application may provide similar facilities in a more convenient form. ff you are not sure, refer to the application's user instructions or ask your technical support team.

WARNING: NEVER connect the AC adaptor with alkaline batteries installed and charging enabled! They will NOT re-charge. Instead. they may explode, jam in the battery tube or leak corrosive chemicals.

6.5.6 AC Operation

THESE INSTRUCTIONS RELATE TO PERSONNEL SAFETY AS WELL AS RELIABLE OPERATION OF HUNTER 16 AND AC ADAPTOR. IT IS IMPORTANT THAT THEY ARE READ AND UNDERSTOOD.

Hunter 16 can be powered permanently from the AC adaptor by fitting NiCads and leaving the unit connected.

For critical or remote operations, charging should be authorised. The batteries will then remain "topped up', so that the unit can continue operating through interruptions to AC power such as a power cut. This is acceptable provided ventilation around the AC adaptor and **Hunter 16** is good, but ageing of the cells may be accelerated. Annual replacement would be a prudent action in this instance.

This will also mean that the unit can be used as a portable at any time, just by disconnecting the AC adaptor.

6.5.6.1 Connecting the AC adaptor

To use the AC adaptor the following procedure is recommended:

1. Switch off **Hunter 16**.
2. If Alkaline batteries are fitted, fit NiCads instead, as described in section 6.5.4.3.
3. Plug the AC adaptor into **Hunter 16's** 5-way connector, by aligning the red mark on the plug sleeve with the corresponding red mark on the socket. Push the plug in until the outer sleeve is flush with the **Hunter 16** casing.
4. Plug the AC adaptor into the AC.

5. Switch on the AC socket.
6. Switch on **Hunter 16** and use it as normal.
7. When you want to disconnect, turn off the AC, then disconnect the AC adaptor by removing the plug.

NOTE: Only 240V (220 to 250V) 50 or 60 Hz AC is suitable for standard units. Adaptors are also available for alternative supplies such as 110V.

While operating from the AC adaptor, the minimum battery level returned will be 6%. If charging is enabled, the NiCads will recharge.

6.6 Everyday maintenance

Hunter 16 needs very little routine maintenance.

Its LCD screen and case can be cleaned with a lens cloth or damp chamois leather. Do not use solvent cleaners.

The battery plug relies on an '0 ring for case sealing. If sealing is important (eg **Hunter 16** is being used regularly in wet conditions), you should inspect the silicon grease on this ring at regular intervals and renew it if required.

If batteries are fitted, they should be replaced or recharged at regular intervals, as described in section 6.5.

6.7 Extending, repairs and internal upgrades

If anything should go wrong with **Hunter 16**, it is of course covered by a warranty. For the terms and details of this warranty, contact your supplier.

Repairs should only be performed by Husky Computers Ltd or your local distributor. In the first instance, contact the distributor specified on the first page of this guide.

You should also contact Husky Computers Ltd or your local distributor if you want to extend Hunter 16's facilities by adding extra internal devices.